Kerr Landau-Siegel proof evaluation, Step 4: citation check against the originals

Document evaluated: Bryce Kerr, "A simpler proof that there are no Landau–Siegel zeros," arXiv:2610.21099v1 [math.NT], 15 October 2026.

Built on: Steps 1, 2 and 3: - "Step 1 - Kerr Landau-Siegel proof evaluation Step 1 18 Oct 2026 1708_v1.md" - "Step 2 - Kerr Landau-Siegel proof evaluation Step 2 18 Oct 2026 1746_v1.md" - "Step 3 - Kerr Landau-Siegel proof evaluation Step 3 18 Oct 2026 1752_v1.md"

Sources read in this step:

Kerr ref.	Source	Copy used	What I read
[5]	D. W. Masser, "On polynomials and exponential polynomials in several complex variables," *Inventiones Mathematicae* 63 (1981), 81 to 95	Arnoldo's PDF, 15 pages	In full: Sections 1 to 5 (the statement and complete proof of Theorem 1). Sections 6 and 7 concern Theorem 2, which Kerr does not use. I checked them only for anything bearing on Theorem 1, and found nothing. Pages 82, 83 and 90 were checked against the page images, not only the extracted text.
[2]	H. Davenport, *Multiplicative Number Theory*, 2nd edition, revised by H. L. Montgomery, Graduate Texts in Mathematics 74, Springer, 1980	Arnoldo's images of pp. 88 and 89, and the file "Multiplicative Number Theory 1980.pdf" in "Papers and Books"	Chapter 14, pp. 88 to 90
[4]	J. S. Milne, *Algebraic Number Theory*, course notes	The file "Algebraic Number Theory - J.S. Milne - ANTe6.pdf" in "Papers and Books", and Arnoldo's images of pp. 274 and 275	§8 "Global Fields": the definition of the Frobenius element (p. 270), Examples 8.18 and 8.19, and the Application (pp. 273 to 275)

Question for each citation: does the source state what Kerr uses it for, under hypotheses Kerr's situation satisfies? Anything I derive to bridge a gap is labeled as mine.

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1. Masser [5] against Kerr's Lemma 6 (Step 2, item B22)

1.1 What Masser states

Masser, p. 82 (checked against the page image):

> Let Γ be a non-trivial finitely generated additive subgroup of C^n. Let $l \geq 1$ be its rank, and for $1 \leq r \leq n$ let l_r be the maximum rank of any subgroup Γ' of Γ for which there is a subspace V of C^n of dimension r with $v \cdot \gamma = 0$ for all v in V and all γ in Γ'. Define $\mu(\Gamma) = \min_{1\leq r\leq n} (l - l_r)/r$. Let $\gamma_1, \ldots, \gamma_l$ be **basis elements** of Γ, and let $\Gamma(N)$ be the set of points $n_1\gamma_1 + \cdots + n_l\gamma_l$ with non-negative integers n_i not exceeding N.
>
> **Theorem 1.** For real $N \geq 0$ and $D \geq 1$, let P be a non-zero polynomial of total degree at most D such that $P(\gamma) = 0$ for all γ in $\Gamma(N)$. Then $D \geq (N/n)^{\mu}$.

This is a paraphrase. The symbols are Masser's.

1.2 Comparison with Kerr's Lemma 6

Point	Masser, Theorem 1	Kerr, Lemma 6	Finding
(a) Points	$\gamma_1, \ldots, \gamma_l$ are a **basis** of Γ, so l = rank Γ	$\gamma_1, \ldots, \gamma_s$ are **any generators** of Γ	**Kerr's statement is broader than the source.**
(b) Exponent	$\mu = \min_r (l - l_r)/r$, defined through annihilators	μ = min over proper L of rank$(\Gamma/(\Gamma \cap L))/(n - \dim L)$	**Equivalent** (my proof in 1.3). Kerr never shows this.
(c) Range	Non-negative integers not exceeding N, with $N \geq 0$ real	Integers $0 \leq m_j \leq M$, with $M \geq 1$ an integer	Kerr's is a special case. No issue.
(d) Degree	"Total degree at most D," with $D \geq 1$ real; conclusion $D \geq (N/n)^{\mu}$	Conclusion deg $F \geq (M/n)^{\mu}$	Take D = deg F. This needs deg $F \geq 1$, which holds because a nonzero constant cannot vanish at the point 0 (all $m_j = 0$). Kerr leaves this unsaid. Routine.
(e) Non-triviality	Γ non-trivial	Not assumed	If every $\gamma_j = 0$, then $\mu = 0$ and the bound deg $F \geq 1$ holds trivially. No

			issue.
(f) Scalar product	The bilinear form $x \cdot y = \Sigma x_i y_i$ (as Masser's proof, p. 91, uses it)	Not needed	Used only in the equivalence of 1.3.

1.3 The two definitions of μ agree (my derivation)

Under the bilinear form $x \cdot y$, which is nondegenerate, the subspaces V of dimension r correspond one to one with the subspaces $L = V^\perp$ of dimension $n - r$.

A subgroup $\Gamma' \subseteq \Gamma$ satisfies $v \cdot \gamma = 0$ for all $v \in V$ and $\gamma \in \Gamma'$ exactly when $\Gamma' \subseteq L$. The largest such subgroup is $\Gamma \cap L$. Hence l_r = max over subspaces L with $\dim L = n - r$ of $\operatorname{rank}(\Gamma \cap L)$.

Also $\operatorname{rank}(\Gamma/(\Gamma \cap L)) = l - \operatorname{rank}(\Gamma \cap L)$, because ranks add along the exact sequence $0 \to \Gamma \cap L \to \Gamma \to \Gamma/(\Gamma \cap L) \to 0$. Put $r = n - \dim L$, which runs over 1, …, n as L runs over the proper subspaces. Then Kerr's μ is $\min_r \min_{\{\dim L = n-r\}} (l - \operatorname{rank}(\Gamma \cap L))/r = \min_r (l - l_r)/r$, which is Masser's μ. ∎

1.4 Is Kerr's Lemma 6 true with arbitrary generators? (my derivation)

Yes. It follows from Masser's Proposition (§2, p. 83, checked against the page image) by rerunning his §5 proof with s generators in place of a basis. Masser's paper does not state this.

Masser's Proposition (paraphrase). Let $E_1, \ldots, E_h$ be commuting affine automorphisms of $K[x_1, \ldots, x_n]$, and let $\mathfrak{M}$ be a proper maximal ideal. For $1 \le r \le n$, let h_r be the largest rank of a stabilizer $S(\mathfrak{P}) \subseteq Z^h$, as $\mathfrak{P}$ runs over the prime ideals of rank r contained in $\mathfrak{M}$. Suppose $h_r < h$ for all r, and let N, $D \ge 1$ be real numbers with $N \ge D^{\eta_1} + \cdots + D^{\eta_n}$, where $\eta_r = r/(h - h_r)$. Then for any non-zero P of total degree at most D, some translate $E_1^{n_1} \cdots E_h^{n_h} P$ with $0 \le n_i \le N$ does not lie in $\mathfrak{M}$.

The derivation. 1. Take $h = s$ and E_i = translation by γ_i ($i = 1, \ldots, s$), with $\mathfrak{M} = (x_1, \ldots, x_n)$, exactly as on p. 90 but with generators instead of a basis. 2. Let $\varphi: Z^s \to \Gamma$ be $\varphi(v) = \Sigma v_i \gamma_i$. It is surjective, and its kernel has rank $s - l$. The translate E^v is translation by $\varphi(v)$. So the stabilizer of a prime $\mathfrak{P}$ is $S(\mathfrak{P}) = \varphi^{-1}(\Gamma')$, where $\Gamma' = \{\gamma \in \Gamma :$ translation by γ fixes $\mathfrak{P}\}$. Since $\ker \varphi \subseteq S(\mathfrak{P})$, $\operatorname{rank} S(\mathfrak{P}) = \operatorname{rank} \Gamma' + (s - l)$. 3. Masser's proof of his inequality (10), on pp. 90 and 91, shows $\operatorname{rank} \Gamma' \le l_r$ for every prime $\mathfrak{P} \subseteq \mathfrak{M}$ of rank r. That argument uses only Γ' and the prime $\mathfrak{P}$, not the choice of generators. Hence $h_r \le l_r + (s - l)$, so $h - h_r \ge l - l_r$. 4. If $\mu > 0$, then $l_r < l$ for all r, so $h_r < h$, and $\eta_r = r/(h - h_r) \le r/(l - l_r) \le 1/\mu$. If F vanishes at every $\Sigma m_j \gamma_j$ with $0 \le m_j \le M$, all the translates vanish at the origin. The Proposition then forces $M < D^{\eta_1} + \cdots + D^{\eta_n} \le n D^{1/\mu}$ (when $M \ge 1$). With $D = \deg F$ this gives $\deg F > (M/n)^\mu$. 5. If $\mu = 0$, the claim is $\deg F \ge 1$, which holds as in 1.2(d). ∎

1.5 Does Kerr's application satisfy Masser's hypotheses as stated?

Yes, directly. - $\Gamma = \pi(Z^4)$ has rank 4. Lemma 7's proof shows π is injective on Z^4, which holds when $q \ne 8$ (Step 2, C.10). - So $\gamma_1, \ldots, \gamma_4$ are basis elements of Γ, and the points $n_1\gamma_1 + \cdots + n_4\gamma_4$ with $0 \le n_j \le N - 1$ are exactly Masser's $\Gamma(N - 1)$. - $n = 3$, and $\deg F \ge 1$ by 1.2(d). Masser's Theorem 1 applies as stated, with no need for 1.4.

1.6 Verdict on B22

- Kerr's Lemma 6 is **true**: it agrees with Masser's Theorem 1 for bases (1.3), and holds for generators by my derivation (1.4).
- The **citation does not cover it as written.** Masser assumes a basis. The generator form needs the extra argument of 1.4, which is not in Masser and not in Kerr.
- The **application is covered** by Theorem 1 exactly as Masser states it (1.5). The page does not say this, and does not point out that rank Γ = 4 is what makes $\gamma_1, \ldots, \gamma_4$ a basis.
- There is no false step. B22 remains a break, and is **downgraded from substantive to routine**: the import that carries the argument is sound, and the gap is in how it is cited.

One further observation, not a break in Kerr. Masser's §5 uses the Proposition with $\eta_r \le 1/\mu$ and then $N < nD^{1/\mu}$. Equality cases do not arise. Masser's inequality signs on p. 90 were checked against the image.

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2. Davenport [2] against Kerr's Lemma 1 (Step 2, items B4 and C.1)

2.1 Edition and location

The 1980 second edition (the "Multiplicative Number Theory 1980.pdf" file, title page: "Second Edition, Revised by Hugh L. Montgomery") has Chapter 14, "Zero-Free Regions for L(s, χ)", beginning on p. 88. Equation (2) is on p. 89. **The text matches Arnoldo's images word for word, and Kerr's citation "[2, Equation (2), p. 89]" is accurate as to edition, page and equation.**

2.2 What p. 89 states, and the hypotheses

- **Equation (2):** $-\mathrm{Re}\, L'/L(s,\chi) < c_2\mathscr{L} - \Sigma_\rho \mathrm{Re}\, 1/(s-\rho)$, where $\mathscr{L} = \log q + \log(t+2)$ by Equation (3). The text says (2) "holds (for $\sigma > 1$) for any primitive χ, whether real or complex".
- **The standing hypotheses of Chapter 14 (p. 88):** "We no longer suppose that $t \geq 2$ but merely that $t \geq 0$", and "We are concerned with nonprincipal characters only, and therefore $q \geq 3$ throughout."
- **Positivity (p. 89):** $\mathrm{Re}\, 1/(s-\rho) = (\sigma-\beta)/|s-\rho|^2 \geq 0$, so "we can as before omit the series or any part of it."
- **The unnumbered display (p. 89):** $-L'/L(\sigma,\chi_0) \leq -\zeta'/\zeta(\sigma) < 1/(\sigma-1) + c_1$ for $1 < \sigma < 2$, where c_1 "denotes a positive absolute constant (and similarly for c_2, ... later)."

2.3 Does it support Lemma 1 as used?

Yes, together with the unnumbered display, which Kerr does not cite (F15). My derivation C.1 uses exactly these four facts, at $t = 0$, $\sigma = 1 + 1/\log X \in (1, 2)$, with $\chi = (q/\cdot)$ primitive of conductor $|q| \geq 3$ (F13), keeping the single term $\rho = \beta$.

Davenport himself makes the same move on p. 90: "by retaining on the right of (2) only the one term corresponding to this zero."

Verdict on B4: the source fully supports the lemma. The break is that the derivation, the second display, and the conductor fact are not on Kerr's page. B4 remains a break. It stays substantive because it is the analytic input of the whole argument, though the derivation is short (C.1).

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3. Milne [4] against Kerr's Frobenius congruences (Step 2, items B7 and C.3)

3.1 Version and location

The ANTe6 file is **Milne, *Algebraic Number Theory*, Version 3.03 (May 29, 2011)**, typeset in a 327-page small-page format. In it, §8 "Global Fields" begins on p. 257. "The Frobenius element" is on p. 270, and Example 8.19 is on **p. 274**. **Arnoldo's images of pp. 274 and 275 come from this file.**

Kerr's reference gives no version, only "course notes, available online". The section number "§8" matches v3.03. I have not checked other versions, where the numbering could differ.

3.2 What Example 8.19 states

> Let $K = \mathbb{Q}[\sqrt{d}]$, and let p be a prime that is unramified in K. Identify $\mathrm{Gal}(K/\mathbb{Q})$ with $\{\pm 1\}$. Then $(p, K/\mathbb{Q}) = +1$ or -1 according as p does, or does not, split in K, i.e., according as d is, or is not, a square modulo p. Thus $(p, K/\mathbb{Q}) = (d/p)$.

The Frobenius element (pp. 270 and 271) is the unique σ in the decomposition group $G(\mathfrak{P})$ with $\sigma\alpha \equiv \alpha^{N\mathfrak{p}} \bmod \mathfrak{P}$ for all α in the ring of integers O_L, where $\mathfrak{P}$ is a prime of L unramified over $\mathfrak{p}$ and $N\mathfrak{p}$ is the size of the residue field of $\mathfrak{p}$.

3.3 Does it support Kerr's congruences?

Kerr uses $(\sqrt{q})^p \equiv \chi(p)\sqrt{q}$ and $(\sqrt{2})^p \equiv (2/p)\sqrt{2}$ **modulo pO**, where $O = \mathbb{Z}[\sqrt{q}, \sqrt{2}]$. Example 8.19 gives an element of the Galois group of $\mathbb{Q}(\sqrt{d})/\mathbb{Q}$. Getting from it to Kerr's congruences takes four steps, none of them on Kerr's page or in Milne's example:

1. Apply the definition (pp. 270 and 271) to $\alpha = \sqrt{d}$, for $d = q$ and $d = 2$ separately. This gives $(\sqrt{d})^p \equiv (d/p)\sqrt{d}$ modulo each prime $\mathfrak{P}$ over p in $O_{\mathbb{Q}(\sqrt{d})}$.
2. Since p is unramified, $pO_{\mathbb{Q}(\sqrt{d})}$ is the product of the distinct primes over it. So the congruence holds modulo $pO_{\mathbb{Q}(\sqrt{d})}$, hence modulo pO_K.
3. Pass from pO_K to pO. This needs $p \nmid [O_K : O]$. **My check:** the discriminant of the basis $1, \sqrt{q}, \sqrt{2}, \sqrt{(2q)}$ is $(32q)^2$, because the embedding matrix is the Kronecker product of $[[1, \sqrt{q}], [1, -\sqrt{q}]]$ and $[[1, \sqrt{2}], [1, -\sqrt{2}]]$, with determinant $(-2\sqrt{q})^2(-2\sqrt{2})^2 = 32q$. So $[O_K : O]$ divides $32|q|$, and an odd $p \nmid q$ does not divide it. Then $O \cap pO_K = pO$: if $x = py$ with $y \in O_K$, choose integers a, b with $a[O_K : O] + bp = 1$; then $y = a([O_K : O]y) + bx \in O$.
4. Identify Milne's Legendre symbol (d/p) with Kerr's Kronecker symbol $\chi(p) = (q/p)$ for odd $p \nmid q$.

Verdict on B7: - The citation is **correct in substance but indirect.** Milne supports the Frobenius element, not the congruence modulo pO that Kerr writes. - Bridging the two needs the four steps above, including a ring-index argument that Kerr never mentions. - The direct route through Euler's criterion (my derivation C.3) takes one line and needs no Frobenius theory. - B7 remains a routine break. There is no false step.

Aside (Milne, not Kerr). On p. 275 Milne's display reads $(-1/p) = 1$ if $p \equiv 1 \bmod 4$ and -1 if $p \equiv 1 \bmod 4$. The second case should read $p \equiv 3 \bmod 4$. This is a typographical error in the Application to quadratic reciprocity, which Kerr does not use. It does not affect Example 8.19.

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4. Effect on the Step 2 tally

Item	Step 2	After Step 4	Reason
B22 (Lemma 6)	Break, substantive	**Break, routine**	The statement is true (1.3, 1.4), and the application meets Masser's Theorem 1 as stated (1.5). What is missing is citation precision and the remark that rank $\Gamma = 4$.
B4 (Lemma 1)	Break, substantive	Break, substantive (unchanged)	The source confirms the lemma, but the derivation is still off the page.
B7 (Frobenius)	Break, routine	Break, routine (unchanged)	The source supports less than Kerr asserts. Bridging it takes four steps.
FS1 (q = 8)	False step	False step (unchanged)	Not affected by the sources.

Revised tally: 1 false step and 32 breaks, now **7 substantive** (B1, B2, B4, B20, B21, B26, B28) and **25 routine**, plus 3 notation defects. The 7 substantive breaks are all missing derivations or missing structure. None is a doubt about whether the claim is true: Step 2 Part C supplies each one, and Step 3 confirms every number.

Additions to the foundation inventory (beyond F13 to F15 from Step 2): - **F16:** the passage from Masser's Theorem 1 to Lemma 6 as written needs either a basis (Kerr's application has one) or Masser's Proposition (§2) with the argument of 1.4. - **F17:** the ring index $[O_K : O]$ divides $32|q|$. Needed only if the Frobenius route of 3.3 is used instead of Euler's criterion.

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5. Where the evaluation stands after Steps 1 to 4

On the page, the paper is **not a complete proof** by our standard. Its foundation is unstated, and seven substantive steps are left to the reader. One statement is false (q = 8).

Above the foundation, **I found no defect in the mathematics.** Every gap can be filled, and Step 2 Part C, together with Sections 1.3, 1.4 and 3.3 here, fills each one. Every number is confirmed by machine (Step 3), and every citation supports what it is used for, once the bridging steps are supplied (Step 4).

That is a statement about what has been checked. It is not a claim that the theorem is established beyond question. Step 5 takes up the plausibility, effectivity, and outside-reaction questions.

Proposed Step 5: plausibility and context, labeled by band

This step is judgment, kept separate from the on-page findings. Every claim will carry a band label: **Band 1** (shown in Steps 1 to 4), **Band 2** (my reading, for an expert to check), or **Band 3** (heuristic).

1. **"Proves too much" tests.**
    - a. List what the theorem would imply, and check each against known results and computations.
    - b. Check whether the argument depends on the target being a genuine L-function. For example, does it apply to functions known to have real zeros near 1?
    - c. Find exactly where the Euler product, or the Frobenius structure, enters.
2. **Effectivity.** Can c be computed? This depends on the constants C_1 (Davenport's c_1 and c_2), C_M (Mertens), Chebyshev's constant, the threshold N_0 (Step 3 suggests it is small), and a finite computation for small $|q|$. An effective c would give an effective Siegel theorem, and so effective class number bounds. Kerr does not state this.
3. **How the method works.** Why the earlier barriers to excluding a single exceptional zero (Siegel's ineffectivity, and the limits of zero-density and sieve methods) do not block this argument. How it relates to the OpenAI manuscript and to Banks's exposition, both of which you supplied. I would read both in full for the comparison. How it relates to our Quasi-Riemann Hypothesis evaluation, which also claimed to exclude Siegel zeros.
4. **Outside reaction to Kerr, Banks and the OpenAI paper.** This needs a web search, which returns tool summaries rather than original texts. I will say so wherever I rely on one, and read in full any source you want treated as authoritative.
5. **Output:** "Step 5 - Kerr Landau-Siegel proof evaluation Step 5 18 Oct 2026 HHMM_v1.md", saved in the folder.